NIFAS SILK LAFTO SUB CITY EDUCATION OFFICE. ADDIS ABABA

FIRST SEMESTER GRADE 12 BIOLOGY MODEL EXAMINATION.

TIR 2018/JANUARY 2026

NUMBER OF QUESTIONS: 100

TIME ALLOWED: - 2 HOURS

GENERAL DIRECTIONS

THIS BOOKLET CONTAINS BIOLOGY EXAMINATION. IN THIS EXAMINATION. THERE ARE A TOTAL OF 100 MULTIPLE CHOICE QUESTIONS. CAREFULLY SELECT THE BEST ANSWER AND BLACKEN ONLY THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED. FOLLOW THE INSTRUCTIONS ON THE ANSWER SHEET AND THE EXAMINATION PAPER CAREFULLY. USE ONLY PENCIL TO MARK YOUR ANSWERS. YOUR ANSWER MARK SHOULD BE HEAVY AND DARK. COVERING THE ANSWER SPACE COMPLETELY. PLEASE ERASE ALL UNNECESSARY MARKS COMPLETELY FROM YOUR ANSWER SHEET.

YOU ARE ALLOWED TO WORK ON THE EXAM FOR 2 HOURS. WHEN TIME IS CALLED. YOU MUST IMMEDIATELY STOP WORKING. PUT YOUR PENCIL DOWN. AND WAIT FOR FURTHER INSTRUCTIONS.

ANY FORM OF CHEATING OR AN ATTEMPT TO CHEAT IN THE EXAMINATION WILL RESULT IN AN AUTOMATIC DISMISSAL FROM THE EXAMINATION HALL AND CANCELLATION OF YOUR SCORE (S..

PLEASE MAKE SURE THAT YOU HAVE WRITTEN ALL THE REQUIRED INFORMATION ON THE ANSWER SHEET BEFORE YOU START TO WORK ON THE EXAMINATION.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

1. Which of the following is classified as a non-renewable natural resource?

A. Timber B. Coffee plants C. Gold deposits D. Sunlight

2. Which of the following best explains why a forest, which is generally considered renewable, may be lost permanently?

A. It requires too much sunlight for regeneration

B. Overharvesting could lead to extinction

C. It competes with non-renewable resources like minerals

D. It is converted into fossil fuels over time

3. Which of the following best analyzes the differences between food security and nutrition security?

A. Food security and nutrition security are identical, both meaning sufficient calories.

B. Food security focuses on access to sufficient food, while nutrition security on meeting dietary needs.

C. Nutrition security is just a subset of food security, limited to micronutrients.

D. Food security is only about economic access to food and nutrition security depends solely on health services.

4. How do ecological concepts mainly help to create conscious citizens? By

A. building awareness of human impacts on ecosystems to promote responsible actions.

B. developing advanced genetic engineering skills for crop improvement in agriculture.

C. providing detailed knowledge of human physiology without societal or environmental links.

D. focusing on memorization of classification systems for Ethiopian plants and animals.

5. Which of the following illustrates the application of biological knowledge in promoting personal health and wellbeing through understanding human nutrition?

A. Recognizing the role of balanced nutrients in preventing lifestyle diseases like diabetes

B. Studying animal migration patterns to inform conservation efforts

C. Applying microbiology principles to develop new antibiotics for bacterial infections

D. Using genetic engineering to create new crop varieties for increased yield

6. A student wants to improve personal health and prevent non-communicable diseases like hypertension. Which of the following practices is recommended?

A. Consuming large amounts of protein supplements to build muscle mass quickly

B. Drinking excessive amounts of water to flush out toxins from the body

C. Taking multivitamin pills on daily basis.

D. Maintaining regular physical activity and including proper diet

7. Which of the following characteristics best distinguishes Eubacteria from Archaea and eukaryotic organisms?

A. Presence of a true nucleus surrounded by a nuclear membrane

B. Possession of peptidoglycan in the cell wall

C. Ability to perform photosynthesis using chlorophyll a

D. Presence of linear chromosomes with histone proteins

8. Which of the following physiological features enables many archaea to maintain membrane integrity and function at very high temperatures and extreme pH levels?

A. Cell membranes composed of ester-linked straight-chain fatty acids like eubacteria

B. Cell membranes with ether-linked branched isoprenoid lipids that resist hydrolysis.

C. Presence of peptidoglycan in the cell wall that provides rigidity in high-salt environments

D. Use of chlorophyll-based photosynthesis to generate energy in anaerobic conditions

9. What is the primary ecological role of saprophytic fungi in nutrient cycling?

A. Breaking down dead organic matter to release nutrients back into the soil

B. Producing oxygen through photosynthesis like plants

C. Fixing atmospheric nitrogen into usable forms for plants

D. Providing shelter for small animals in forest habitats

10. Amoebic dysentery which is caused by the protozoan *Entamoeba histolytica*, is commonly transmitted through which mode of transmission?

A. Inhalation of airborne cysts from dried fecal material

B. Bite of a tsetse fly vector in endemic regions

C. Fecal-oral route via ingestion of contaminated food or water

D. Sexual transmission through direct mucosal contact

11.Viruses are considered unique among biological entities primarily because of which of the following general characteristics?

A. They possess both DNA and RNA as genetic material simultaneously

B. They have a true nucleus and membrane-bound organelles like eukaryotic cells

C. They are obligate intracellular parasites that can only replicate inside living host cells

D. They can perform independent metabolism and produce energy outside a host

12. Which of the following dietary practices most significantly promote a healthy and diverse gut microbiota?

A. A high-protein, low-carbohydrate diet that minimizes fiber intake

B. Frequent consumption of sweeteners that selectively inhibit harmful bacteria

C. A diet rich in dietary fiber and plant-based foods.

D. Exclusive reliance on processed foods.

13. In the nitrogen cycle, certain microorganisms convert atmospheric nitrogen into a form usable by plants. This process demonstrates how microorganisms facilitate nutrient recycling by:

A. Transforming inert nitrogen gas into ammonia or nitrates for plant uptake.

B. Breaking down organic waste into carbon dioxide.

C. Releasing phosphorus from rocks into soil.

D. Oxidizing sulfur compounds to produce energy.

14. Antibiotics like penicillin are used to treat bacterial infections in patients by:

A. Preventing cell wall synthesis.

B. Inhibiting integration of endoplasmic reticulum.

C. Blocking DNA replication in all microorganisms.

D. Lowering the temperature around bacterial cells.

15. Gram staining is a differential staining technique used to classify bacteria into two major groups. The main basis for this classification is:

A. The shape of the bacterial cells. C. The size of the bacteria under the microscope.

B. The ability of bacteria to form spores D. Differences in the structure of cell wall.

16. Which Ethiopian scientist is renowned for discovering medicine from the 'Endod' plant to control schistosomiasis, a parasitic disease transmitted by snails?

A. Gebisa Ejeta B. Segenet Kelemu C. Aklilu Lemma D. Belay Tessema

17. Cellular respiration is primarily a catabolic pathway because it:

A. Builds large molecules from smaller subunits using energy.

B. Uses light energy to produce glucose

C. Synthesizes proteins directly from carbon dioxide.

D. Breaks down glucose and releases energy stored in ATP.

18. Plants grown under green light appear less healthy and grow slower than those under red or blue light because;

A. Chlorophyll reflects green light most during the process of photosynthesis.

B. Chlorophyll absorbs green light strongly, preventing it from reaching reaction centers.

C. Chlorophyll directly converts green light into chemical energy.

D. The chloroplast stroma reflects all visible light equally.

19.The primary photosynthetic pigment found in all photosynthetic organisms is:

A. Chlorophyll b B. Chlorophyll c C. Chlorophyll a D. Chlorophyll d

20.The efficiency of photosynthesis in C4 plants outperform C3 plants mainly because: They

A. have lower rates of dark respiration at night.

B. lack grana, preventing oxygen evolution.

C. fix CO_2 directly into a three-carbon compound, bypassing Rubisco.

D. avoid energy loss in the form photorespiration.

21. C4 metabolism in plants is more advantageous in environments with:

A. High light intensity, high temperature, and low water availability.

B. Cool, moist and shaded habitats.

C. Conditions with very low CO_2 concentrations.

D. All environments equally.

22. Which of the following best describes the direct relationship between the thylakoid membrane and the stroma activities? The

A. thylakoid reactions provide ATP and reduced NADP to the stroma.

B. stroma provides CO2 to the thylakoid to initiate photolysis.

C. thylakoid reactions produce G3P which is then converted to ATP in the stroma.

D. stroma produces oxygen which is then used by the thylakoid membrane.

23. Photosynthesis is considered the most fundamental process for the continuity of life on Earth because it:

A. regulates the temperature of the internal environment of plants.

B. converts radiant energy into chemical energy stored in organic molecules.

C. absorbs oxygen from the atmosphere to facilitate cellular respiration.

D. produces carbon dioxide which is essential for the survival of heterotrophs.

24.What is the net gain of ATP molecules produced from one molecule of glucose during anaerobic respiration?

A. 36 to 38 ATP B.0 ATP C.4 ATP D.2 ATP

25. Which of the following cells would likely contain the highest density of mitochondria?

A. Skin cell B. Cardiac muscle cell C. Fat storage cell D. Mature red blood cell

26. During the Krebs cycle or Citric Acid Cycle, which of the following is NOT produced?

A. C02 B. FADH C. NADH D. O2

27. In the electron transport chain, what is the final electron acceptor?

A. NAD+ B. Water C. Oxygen D. Carbon dioxide

28. In the production of traditional Ethiopian 'Tella', which of the following best explains the role of 'Bikil' (malted grain)? It:

A. provides the enzymes necessary to hydrolyze starch into fermentable sugars.

B. acts as the primary anaerobic electron acceptor for ethanol production.

C. serves as a preservative to inhibit the growth of acetic acid bacteria.

D. is the main source of carbon dioxide in the final bottled product.

29. During the fermentation of 'Tej', if the container is not properly sealed and oxygen enters. What is the most likely outcome?

A. Glucose will be converted directly into carbon dioxide and water without alcohol.

B. The ethanol will be oxidized into acetic acid by aerobic bacteria.

C. The rate of fermentation will increase due to higher ATP yield.

D. Lactic acid will be produced instead of ethanol.

30. Compare the energy efficiency of glucose metabolism in a person performing a 100m sprint versus a person walking slowly. Which statement is correct?

A. The sprinter produces more CO2 per glucose molecule than the walker.

B. Both individuals produce the same amount of net ATP per glucose molecule.

C. The walker utilizes NADH to reduce pyruvate into lactate.

D. The sprinter produces less ATP per glucose molecule due to the accumulation of lactate.

31. The branch of biology that deals with the study of interactions of living organisms with their environment is called:

A. Ecology B. Microbiology C. Genetics D. Physiology

32. Which of the following branches of biology is considered a 'Pure Biology' field?

A. Taxonomy B. Biotechnology C. Pharmacology D. Agronomy

33. The discovery of the double-helix structure of DNA by Watson Crick and Franklin fundamentally changed biology by:

A. explaining the mechanism of natural selection

B. proving that cells come from pre-existing cells

C. revealing how genetic information is stored and copied

D. developing the first vaccine for rabies

34. Plants carry out all of the following life processes. except:

A. growth B. reproduction C. nutrition D. motility

35. In the classification of Angiosperms, what is the primary distinction between Monocots and Dicots regarding their floral parts?

A. Monocots typically have floral parts in multiples of three.

B. Monocots lack sepals and petals.

C. Dicots always have fewer stamens than Monocots.

D. Monocots have floral parts in multiples of four or five.

36. Which of the following is an advantage of cross-pollination over self-pollination? It

A. ensures pollination in the absence of external agents.

B. preserves the exact genetic characteristics of the parent.

C. leads to greater genetic variation in the offspring.

D. requires less energy expenditure for the plant.

37. Which layer of the leaf contains the highest concentration of chloroplasts and is the primary site for photosynthesis?

A. Upper epidermis B. Palisade mesophyll C. Spongy mesophyll D. Vascular bundle

38. Why is oxygen an essential requirement for the germination of most seeds? To

A. help in the absorption of minerals from the soil

B. oxidize the seed coat so it can break easily

C. synthesize chlorophyll for photosynthesis

D. facilitate aerobic respiration for ATP production

39.A seed characterized by having ‘parachute' of fine hairs is most likely adapted for dispersal by:

A. Animals B. Explosion C. Wind D. Water

40. Which of the following serves as the primary source of carbon for the synthesis of glucose during photosynthesis?

A. Carbon Dioxide B. Water C.ATP D. Soil minerals

41.Which specific enzyme is responsible for the 'fixation' of carbon dioxide by attaching it to a five-carbon sugar at the start of the Calvin Cycle?

A. NADP+ Reductase B. Amylase C. RuBisCO D.ATP Synthase

42. How does photosynthesis contribute to the reduction of the 'Global Warming' effect? By:

A. reflecting sunlight back into space using chlorophyll.

B. producing oxygen, which cools the Earth's surface.

C. acting as a carbon sink, absorbing CO_2

43. What is the primary direction of transport for water in the xylem compared to organic matter in the phloem?

A. Xylem is unidirectional; Phloem is bidirectional.

B. Xylem is bidirectional; Phloem is unidirectional.

C. Both move strictly in a downward direction.

D. Both move strictly in an upward direction.

44. If a germinating seed is placed in a dark box with a single small hole on one side, after a few days, the shoot grows toward the hole. This response is known as:

A. Negative phototropism B. Positive geotropism C. Positive phototropism D.Thigmotropism

45. The amount of energy required to raise the temperature of 1 gram of water by 1°C is very high. This property helps organisms:

A. Maintain a stable internal body temperature. B. Dissolve large amounts of glucose quickly.

C. Transport water to the top of tall trees. D. Evaporate sweat with minimal energy loss.

46. Which electrolyte is primarily responsible for maintaining osmotic balance and nerve impulse conduction in the extracellular fluid?

A. Fluoride B. Sodium C. Potassium D. Zinc

47. A person living in a high-altitude region of Ethiopia might experience swelling in the neck. This condition is most likely caused by a diet deficient in which inorganic element?

A. Iron B. Calcium C. Fluorine D. Iodine

48. Which phase of the cell cycle is characterized by the replication of DNA?

A.S phase B. G1 phase C. Prophase D. G2 phase

49.What is the primary biological importance of meiosis in sexually reproducing organisms? To

A. increases the rate of cell division during embryonic development.

B. double the amount of DNA in the daughter cells.

C. produce genetically identical daughter cells for tissue repair.

D. maintain a constant chromosome number from generation to generation.

50. Which of the following describes the key difference between Anaphase I and Anaphase II of meiosis?

A. In Anaphase I sister chromatids separate, while in Anaphase II homologous chromosomes separate.

B. Anaphase I produces four cells, while Anaphase II produces two cells.

C. In Anaphase I homologous chromosomes separate, while in Anaphase II sister chromatids separate

D.DNA replication occurs between Anaphase I and Anaphase II.

51.Which of the following contribution best represents the role of renowned Ethiopian biologists in science to their society?

A. Importing foreign species for research

B. Identifying, classifying, and conserving endemic species

C. Producing laboratory chemicals

D. developing medical equipment

52. Ethiopian research institutes focus on crop improvement, disease control and biodiversity conservation. Which combination describes the main focus of biological research of these institutions?

A. Space science and astronomy B. Agriculture, health and environmental biology

C. Petroleum engineering and mining D. Mechanical and electrical engineering

53. A laboratory technician attempts to observe viruses using a compound light microscope but fails to see them clearly. Which explanation justifies this limitation?

A. Viruses absorb too much light

B. Viruses are visible only in living cells

C. Viruses require higher resolution than light microscopes provide

D. Viruses lack cell walls

54. A specimen appears too bright, reducing image clarity. The student needs to regulate the amount of light passing through the specimen. Which structure should be adjusted to solve this problem?

A. Arm B. Diaphragm C. Nosepiece D. Stage clip

55.A cell is placed in an environment containing harmful chemicals. Only some substances enter the cell while others are blocked. Which structure is primarily responsible for this selective permeability?

A. Cell wall B. Cytoplasm C. Cell membrane D. Nucleus

56. Red blood cells placed in distilled water swell and may burst, while those placed in concentrated salt solution shrink. Which process explains the movement of water across the cell membrane in both situations?

A. Diffusion of solutes B. Active transport C. Osmosis D. Endocytosis

57. A wilted plant becomes firm after being watered. This change occurs because water enters the plant cells, increasing internal pressure. Why is this process biologically important to plants?

A. It speeds up respiration C. It prevents enzyme activity

B. It enables mechanical support D. It removes metabolic wastes

58. Athletes consume diets rich in rice and bread before competition. Which nutrient is the athlete mainly increasing to meet immediate energy needs?

A. Proteins B. Vitamins C. Carbohydrates D. Minerals

59. Assume an individual takes sugar rich foods regularly to gain energy but later develops health problems. Which statement explains why this diet is unhealthy?

A. Carbohydrates are toxic

B. All nutrients must be included in correct proportions

C. Fats must be completely avoided

D. Vitamins provide energy

60. After digestion, nutrients must enter the bloodstream efficiently. Which digestive organ is structurally adapted for this function with villi?

A. Stomach B. Small intestine C. Large intestine D. Esophagus

61. Digestion begins in the mouth even before food reaches the stomach. Which event explains this early digestion?

A. Protein breakdown by pepsin C. Fat emulsification by bile

B. Starch digestion by salivary amylase D. Glucose absorption by villi

62. A student eats meat, and digestion of proteins starts mainly in the stomach.

Which substance plays the key role?

A. Amylase B. Pepsin C. Lipase D. Maltase

63. Bile does not digest fats directly but is still essential. Why is bile important in fat digestion?

A. It converts fats into glucose C. It neutralizes proteins

B. It emulsifies fats, increasing enzyme efficiency D. It absorbs nutrients

64. Gas exchange occurs efficiently in a structure with thin walls and rich blood supply. Which structure fits this description?

A. Trachea B. Bronchus C. Alveolus D. Larynx

65. During swallowing, breathing temporarily stops to prevent choking. Which structure ensures food does not enter the trachea?

A. Epiglottis B. Bronchiole C. Diaphragm D. Alveoli

66. Oxygen transport decreases when hemoglobin levels are low. Why is hemoglobin essential?

A. It dissolves oxygen in plasma C. It produces oxygen

B. It chemically binds oxygen D. It removes carbon dioxide

67. During inhalation, the diaphragm contracts and flattens. How does this action aid breathing?

A. Decreases chest volume C. Forces air out

B. Increases chest volume, lowering pressure D. Stops gas exchange

68.Cells requires continuous energy for active processes such as muscle contraction and nerve impulse transmission. Which molecule directly supplies this energy?

A. Glucose B. ATP C. Oxygen D. Carbon dioxide

69. After digestion, nutrients are transported to body cells for energy and growth. Which

component performs this transport?

A. Nerves B. Blood C. Hormones D. Enzymes

70. Blood leaving the heart under high pressure travels through vessels with thick muscular walls. Which blood vessel type matches this description?

A. Veins B. Capillaries C. Arteries D. Vena cava

71. Scientists and engineers have learned from nature through imitation of natural facts. Based on this which of the following is incorrectly matched?

A. weaving technology from spider web C. robotic arm from biological neurons

B. air craft the technology from birds D. cutting saw from sharp teeth of animals

72. Among the benefits of biology to technological development, one is not correctly explained?

A. scientists' imitated biological substances in plants to develop artificial version of biochemical substances

B. scientists imitating gene engineering for creation of biosynthetic materials

C. engineers imitated the mound architecture of termites in the design of building.

D. the front of bullet trains has been designed by imitating the flight of birds

73. Which of the following technological device is used to collect bio geographical information?

A. handheld body fat calculator C. geographical position system

B. positron emission tomography D. digital thermometer

74. Which of the following is not among the characteristics of animals?

A. all are multicellular C. all are eukaryotic

B. all have the same body symmetries D. all reproduce

75. Which of the following is not the common feature of invertebrate animals? Invertebrates

A. lack a rigid internal skeletal system C. are the most diverse group of animals

B. do not regulate their body temperature D. includes birds Lizards, amphibians

76 One is not among the characteristics of vertebrate animals? All Vertebrates

A, are advanced group of animal C, have specialized organ systems

B, have internal skeleton system D, are homoeothermic animals

77. In honey bee reproduction the drone is developed from unfertilized egg and the worker bees developed from fertilized eggs. Which type of reproduction is it?

A. Parthenogenesis B. hermaphrodite C. diocious D. monocious

78. The basic difference between complete and incomplete metamorphosis is that complete metamorphosis comprises:

A, young stage B, adult stage C, pupa stage D egg stage

79. Frogs lay eggs in water, and fertilization occurs outside the body. This is an example of:

A. Internal fertilization C. Viviparous reproduction

B. External fertilization D. Asexual reproduction

80. Crocodiles lay shelled eggs on land and protect them until hatching. Which reproductive feature is demonstrated?

A. External fertilization C. Live birth

B. Internal fertilization with egg laying D. Budding

81. Birds incubate eggs to maintain constant temperature. Why is this important?

A. Prevents fertilization C. Stops respiration

B. Ensures embryo development D. Reduces growth

82. Rats give birth to live young that develop inside the mother. This reproductive strategy is called:

A. Oviparity B. Viviparity C. External fertilization D. Fragmentation

83. Silk production depends on silkworms. This shows insects are important to:

A. Tourism B. Industry and economy C. Pollution D. Medicine

84. Birds migrate seasonally to find food and suitable climate. This behavior is BEST classified as:

A. Learned behavior B. Reflex action C. Innate behavior D. Artificial behavior

85. Humans maintain constant body temperature despite environmental change. Such organisms are called:

A. Poikilotherms B. Homeotherms C. Ectotherms D. Cold-blood

86. After eating, blood glucose rises and insulin is released. This illustrates:

A. Digestion B. Blood sugar regulation C. Respiration D. Excretion

87. Sweating during hot conditions helps regulate body temperature. This mechanism maintains:

A. Growth B. Reproduction C. Homeostasis D. Immunity

88. During digestion, starch is rapidly converted into simpler sugars without raising body

temperature. Which property of enzymes explains this efficiency?

A. Enzymes are consumed during reactions

B. Enzymes lower activation energy and remain unchanged

C. Enzymes work only at very high temperatures

D. Enzymes are non-specific

89. An enzyme loses its three-dimensional shape due to extreme pH, even though its amino acid sequence remains unchanged. Which protein structure is MOST disrupted?

A. Primary B. Secondary C. Tertiary D. Peptide bond

90. In a metabolic pathway, excess product slows down its own production by inhibiting the first enzyme. This mechanism is known as:

A. Competitive inhibition B. Feedback inhibition C. Enzyme activation D. Denaturation

91. An enzyme helps join two molecules together using ATP. Which enzyme class performs this function?

A. Isomerase B. Lyase C. Ligase D. Hydrolase

92. A student observes that increasing temperature increases enzyme activity up to a certain point, after which activity drops sharply. What causes the decreasing?

A. Substrate depletion B. Enzyme denaturation C. Increased collisions D. Enzyme activation

93. As substrate concentration increases, enzyme reaction rate rises until it levels off. Why does the reaction reach a maximum rate?

A. Substrate becomes toxic C. Enzyme is denatured

B. All enzyme active sites are occupied D. Product inhibits enzyme

94. Detergents contain enzymes that remove protein stains at low temperatures. Why are enzymes preferred in detergents?

A. They increase fabric damage C. They work efficiently under mild conditions

B. They reduce washing efficiency D. They require boiling water

95. During malting, grains are allowed to germinate before drying. Why is this process important for traditional alcoholic drinks?

A. It destroys starch C. It kills enzymes

B. It converts starch into fermentable sugars D. It prevents fermentation

96 A student compares DNA and RNA and notices differences in structure and function.

Which statement is CORRECT?

A. DNA contains ribose sugar

B. RNA stores hereditary information permanently

C. DNA is double-stranded and stores genetic information

D. RNA is found only in the nucleus

97. Gamete formation reduces chromosome number by half to maintain species stability. Which process and stage are responsible?

A. Mitosis – anaphase C. Meiosis II – reduction division

B. Meiosis I – reduction division D. Mitosis – cytokinesis

98. During protein synthesis, mRNA carries genetic code from nucleus to ribosome. Which role is correctly matched?

A. tRNA – carries genetic code C. DNA – translates proteins

B. rRNA – forms ribosome structure D. mRNA – assembles amino acids

99. A laboratory technician heats an enzyme solution strongly and notices the reaction stops permanently. Which conclusion explains his observation?

A. Enzymes require light C. Enzymes are proteins that denature at high temperature

B. Enzymes are inorganic catalysts D. Enzymes become stronger when heated

100. Which comparison about sexual and asexual reproduction is CORRECT?

A. Sexual reproduction produces identical offspring

B. Asexual reproduction increases variation

C. Sexual reproduction involves fusion of gametes

D. Asexual reproduction requires fertilization